

VoCare AI: A Multi-Agent LLM Workflow for Improved Clinic Operational Efficiency

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Abstract—In Singapore’s polyclinics, touch-based self-service kiosks are widely used for administrative functions such as appointment scheduling and billing. However, these systems pose accessibility challenges for elderly patients, often resulting in increased staff workload and longer wait times. This paper presents *VoCare AI*, a voice-first conversational assistant designed to streamline healthcare administrative tasks through multi-agent workflows powered by Large Language Models (LLMs). The system integrates a graph-based orchestration framework (LangGraph) with specialized agents that manage NRIC-based identity verification, appointment handling, billing queries, and general FAQ responses. It features a fine-tuned Automatic Speech Recognition (ASR) model adapted for Singapore-accented English (Singlish), trained on the IMDA National Speech Corpus to handle accent variability, code-switching, and elderly speech patterns. Evaluation results show that the best-performing ASR model achieved a Word Error Rate (WER) of 22.22%, while the Retrieval-Augmented Generation (RAG) module demonstrated strong performance with 1.00 groundedness and 0.92 retrieval relevance. By focusing on Singapore’s linguistic diversity and incorporating localized authentication mechanisms, this work demonstrates how AI-driven conversational agents can enhance accessibility and operational efficiency in public healthcare environments. The complete source code for this work is publicly available at: <https://github.com/DerrickLJH2000/langgraph-voice-assistant>.

Index Terms—multi-agent LLM, RAG

I. INTRODUCTION

As Singapore’s population continues to age, the healthcare system faces increasing pressure to deliver services that are both efficient and inclusive [1]. To improve operational efficiency, Singapore’s public healthcare sector has adopted digital solutions such as self-service kiosks. An example is the Q’SOFTE Self-Registration Kiosk by ATT Systems Group [2], which enables patients to perform administrative tasks such as appointment scheduling, identity verification, and billing through a touch-based interface. While effective for the digitally literate, these kiosks pose considerable challenges for elderly patients, particularly those with reduced dexterity, vision impairments, or limited familiarity with touchscreen navigation. Such accessibility barriers often lead to increased reliance on clinic staff, longer queues, and operational inefficiencies that offset the intended benefits of digital automation.

Recent advancements in Large Language Models (LLMs) [3], [4] have opened new possibilities for voice-first conversational systems. These models, when integrated with Automatic Speech

Recognition (ASR) [5], [6] and Text-to-Speech (TTS) [7], [8] technologies, enable hands-free, natural language interactions that are particularly valuable in complex service environments such as healthcare. While early adoption of AI-powered voice assistants has been more prevalent in large hospitals and research institutions, there is growing interest in deploying these technologies in small- and medium-sized healthcare enterprises (SMEs) [9], including general practitioner (GP) clinics and community-based polyclinics. In such settings, where staff resources are limited and administrative tasks remain largely manual, LLM-powered assistants can automate appointment scheduling, patient triaging, and information retrieval, thereby improving operational efficiency without extensive infrastructure overhauls.

Voice assistants have shown promise in general domains, and initial explorations in healthcare [10], [11], such as AI-assisted anamnesis and real-time transcription, further demonstrate their potential. However, most existing systems are designed around rigid, rule-based workflows and lack the contextual adaptability needed for handling multi-step, dynamic administrative interactions typical in clinic settings. Similar challenges have been addressed in the hospitality sector, where RAG was integrated with face recognition to enable personalized guest services [12]. These solutions are seldom tailored to the linguistic nuances of local populations. In Singapore, elderly patients frequently converse in Singapore-accented English (Singlish), characterized by localized pronunciation, disfluencies, and spontaneous code-switching between English, Mandarin, Malay, and Tamil. These traits significantly reduce the performance of off-the-shelf ASR models trained primarily on standard English datasets [13], [14]. Moreover, existing systems rarely incorporate orchestrated multi-agent workflows capable of managing real-time user intent recognition, backend service orchestration, and human-in-the-loop intervention, all of which are crucial for reliable deployment in SME clinic contexts.

To address these challenges, we present *VoCare AI*, a voice-first conversational assistant designed for elderly users in Singapore’s polyclinics. *VoCare AI* is built on a multi-agent LLM framework orchestrated using LangGraph, enabling modular and stateful task management across multiple dialogue paths. It incorporates a fine-tuned ASR model trained on Singapore’s IMDA National Speech Corpus [15], capturing

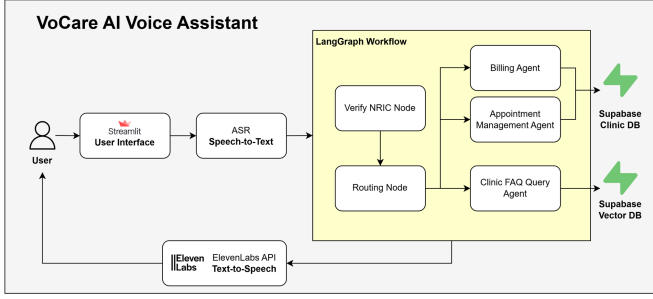


Fig. 1: System architecture of the proposed *VoCare AI* assistant. The system integrates a voice-first interface, a multi-agent conversational workflow orchestrated using LangGraph, and task-specific subgraphs for appointment management, payment processing, and general query handling.

Singlish-specific linguistic patterns. The system supports three core administrative functions, NRIC-based identity verification, appointment scheduling, and payment processing, while also responding to general inquiries through a Retrieval-Augmented Generation (RAG) module grounded in official clinic documents. Our proposed *VoCare AI* has the following contributions:

- We design a multi-agent LLM orchestration workflow using LangGraph to manage complex, stateful conversations for healthcare administrative tasks in a polyclinic setting.
- We adapt and evaluate a Singapore-accented ASR model, fine-tuned on a national speech dataset to robustly handle elderly speech and code-switching patterns.
- We validate the system on curated scenarios, achieving 22.22% Word Error Rate (WER) for ASR and strong RAG performance with 1.00 groundedness and 0.92 retrieval relevance.

II. SYSTEM ARCHITECTURE

In this section, we describe the architecture of our proposed *VoCare AI* by introducing the overall multi-agent design, then explain how user intents are routed through task-specific subgraphs, followed by state management strategies, and finally our adaptation to Singapore’s linguistic environment.

A. Proposed *VoCare AI*’s Multi-Agent Workflow

VoCare AI is designed to mitigate the usability challenges encountered by elderly patients when interacting with traditional touch-based self-service kiosks in Singapore polyclinics. At its core, *VoCare AI* adopts a modular, multi-agent conversational architecture, orchestrated through LangGraph [16]. As illustrated in Fig. 1, the system integrates speech-based input/output, agent-based conversational logic, and a secure backend database.

The multi-agent design follows a formal decomposition of the end-to-end healthcare interaction into specialized sub-processes handled by autonomous agents. Each agent is modeled as a function node in a directed acyclic graph $G = (V, E)$

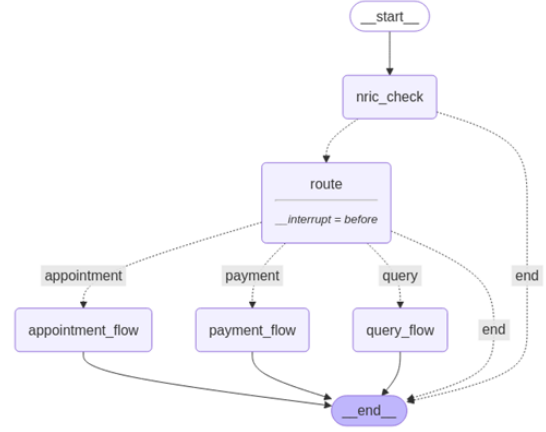


Fig. 2: The conversational flow within *VoCare AI* is initiated by the RootGraph, which analyzes the incoming intent and routes the interaction to either appointment subgraph, payment subgraph, or query subgraph.

consists of a set of nodes V (representing conversational states or functional modules) and edges $E \subseteq V \times V$ (representing transitions based on user intents or task completions). LangGraph manages traversal through this graph based on observed user inputs and intermediate agent outputs, allowing asynchronous, interruptible, and stateful conversation control.

B. Root Graph and Subgraphs

The conversational flow begins with the *RootGraph*, which serves as a global controller. As shown in Fig. 2, once the ASR engine converts spoken input into text, the *RootGraph* processes the intent using a classifier node and routes the flow to the appropriate subgraph. The high-level routes include:

- **Appointment Subgraph:** Handles appointment-related workflows such as querying upcoming appointments, generating time slot recommendations, and confirming rescheduling requests.
- **Payment Subgraph:** Manages voice-driven billing operations, including outstanding charge retrieval, payment confirmations, and receipt generation.
- **Query Subgraph:** Executes open-ended query resolution via RAG that combines vector-based semantic retrieval with LLM-based response synthesis.

Transitions between nodes in each subgraph are implemented using LangGraph’s state machine logic, where state transitions are triggered by user responses or backend signals. To handle context across multi-turn interactions, *VoCare AI* uses *MemorySaver* based on LangGraph’s built-in state checkpointing mechanism. Let s_t denote the graph state at turn t , and h_t be the conversation history. The graph executes transition $s_t \rightarrow s_{t+1}$ conditioned on the tuple (u_t, h_t) , where u_t is the user input. This enables the system to carry forward historical references such as “reschedule that appointment to Monday” or “what was my last charge?”

TABLE I: ASR Model Performance Comparison

Model Name	WER (%)	Evaluation Duration
Whisper-Small Singlish V2 [18]	62.98	59 min 28 sec
Whisper-Base MS Singlish V2 [19]	22.22	32 min 21 sec
Whisper-Small Singlish [20]	67.18	59 min 46 sec

Recognizing the variability in elderly speech, such as slow speech rate, hesitations, or ambiguous phrasing, *VoCare AI* incorporates explicit interrupt points in its LangGraph workflow using the *interrupt_before* parameter. At these interrupt points, the graph execution pauses immediately before designated nodes are executed, explicitly to obtain user input necessary for decision-making or confirmations. This human-in-the-loop approach ensures that critical actions, such as confirming payments or accessing sensitive personal data, are directly verified by the user, enhancing the safety and reliability of system interactions.

C. Adaptation to Singlish through Fine-Tuned ASR

Singapore’s linguistic diversity introduces a key challenge in automatic speech recognition: elderly users frequently code-switch between English and local languages (Mandarin, Malay, Tamil), and exhibit unique pronunciation patterns. To address this, *VoCare AI* employs a Whisper-based ASR model [17] fine-tuned on the IMDA National Speech Corpus (NSC) [15], a corpus curated specifically to reflect conversational speech among Singaporean speakers, including elderly demographics. It includes 2000 hours of transcribed read speech and is enriched with locally relevant words to improve the accuracy of speech engine for Singaporean English accents.

This fine-tuned model improves recognition accuracy by learning lexical features, intonation shifts, and discourse markers common in Singlish (e.g., “lah,” “can anot,” “ah”). Formally, the model learns a mapping $X \rightarrow Y$, where X is the raw audio waveform and Y is the corresponding tokenized text sequence. During inference, beam search decoding is applied over the model’s output logits to generate transcriptions that are robust to code-switching and disfluencies.

III. EXPERIMENTS AND DEMONSTRATION

This section evaluates key components of the *VoCare AI* system, including the performance of its fine-tuned ASR model, the effectiveness of its RAG module, and its end-to-end usability through system implementation and system testing.

A. ASR Model Evaluation

To ensure robust voice-based interaction for elderly users in Singapore polyclinics, we evaluated the performance of ASR models fine-tuned specifically on Singapore-accented English (Singlish), to assess the transcription accuracy for speech patterns commonly observed among elderly speakers. Each model was tested on a curated dataset comprising service-oriented conversations reflective of polyclinic settings. The test set was derived from Part 3 of NSC, which features diverse demographic and linguistic coverage across age groups and

TABLE II: RAG Evaluation Metrics on Clinic FAQ Dataset

Metric	Average Score
Correctness	0.86
Groundedness	1.00
Relevance	0.84
Retrieval Relevance	0.92
Average Latency	4.59 seconds

ethnicities. Word Error Rate (WER), a standard metric that accounts for insertions, deletions, and substitutions relative to the reference transcript, was used for quantitative evaluation. Evaluation duration reflects the total length of the audio processed for each model.

Three Whisper-based ASR models were selected for evaluation: Whisper-Small Singlish V2 [18], Whisper-Base MS Singlish V2 [19], and Whisper-Small Singlish [20]. These models vary in size, training objectives, and the subset of NSC data used during fine-tuning. As shown in Table I, the model developed by Mesolitica achieved the lowest WER at 22.22%, significantly outperforming the others in transcribing elderly Singlish speech; however, the possibility of partial dataset overlap necessitates further validation to ensure generalization to unseen data. These findings highlight the importance of localized fine-tuning when deploying ASR systems in linguistically diverse healthcare environments.

B. RAG Performance and Query Handling

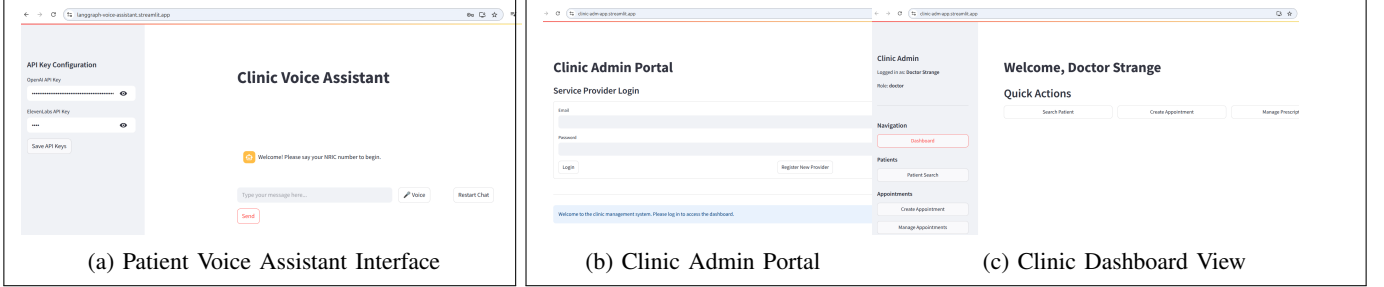
The Query Subgraph in *VoCare AI* leverages a RAG pipeline to answer general clinic-related inquiries. To evaluate its performance, a curated dataset of 50 representative patient questions and their corresponding expected answers was constructed. These queries were drawn from frequently asked questions typically encountered in polyclinic administrative settings. The evaluation was performed using the LangSmith platform, which enabled structured analysis of both the semantic retrieval and the generative components of the RAG pipeline. The results, summarized in Table II, indicate strong performance in both document retrieval and factual response generation, particularly in groundedness and retrieval relevance.

C. System Testing and Evaluation

To ensure the robustness and reliability of the *VoCare AI* system, a comprehensive testing strategy was adopted that addressed both modular components and full-system behavior. The testing process encompassed three key phases: unit testing of individual conversational nodes, integration testing across subgraphs and backend components, and user acceptance testing (UAT) to simulate end-to-end clinical workflows. Unit tests were conducted on each LangGraph node to verify correctness of logic such as NRIC parsing, appointment slot generation, and billing status handling. Special attention was given to edge cases, including malformed inputs, missing session state, and failed backend queries, to ensure that the system could degrade gracefully and provide fallback responses when needed.

TABLE III: Functional Test Cases for *VoCare AI* Workflow

Test Scenario	Test Steps	Expected Result
NRIC Verification	Enter a valid NRIC	System verifies identity and retrieves patient info
Service Routing	State service request (e.g., "appointment")	Routed to correct subgraph (appointment, payment, query)
Appointment Scheduling	Confirm intent to schedule	System proposes 3 available time slots
Timeslot Selection	Select a proposed time slot	Appointment scheduled and updated in database
Payment Retrieval	Request outstanding charges	System retrieves billing details from database
Payment Confirmation	Confirm intent to pay	System updates billing status and issues QR payment prompt
Query Handling	Ask a general clinic question	System responds via RAG with grounded clinic policy info
TTS Response	Trigger system response	Audio reply generated using TTS (ElevenLabs)
STT Input	Speak a command in Singlish	Speech transcribed with fine-tuned ASR and processed

Fig. 3: Screenshots of our *VoCare AI* deployed with Streamlit: (a) voice assistant interface, (b) Admin portal for backend operations, and (c) dashboard interface for clinic staff.

Integration testing focused on the interactions between the RootGraph and its subgraphs (appointment, payment, and query), evaluating the preservation of state and context across multi-turn conversations. This included scenarios involving interrupt handling and memory checkpointing, features essential for elderly-friendly, voice-first experiences. Finally, UAT scenarios were executed on the deployed prototype using seeded patient data in Supabase to validate real-world task execution from the user perspective. As shown in Table III, each test case is mapped to a functional interaction commonly encountered in polyclinics, such as NRIC verification, appointment booking, and payment processing. The system was able to respond accurately to spoken commands in Singlish, route them to the appropriate subgraphs, and update backend records accordingly. These results demonstrate that *VoCare AI* is capable of supporting real clinical workflows with conversational reliability and backend integrity.

D. Implementation and Demonstration

The *VoCare AI* system was implemented as a full-stack prototype to validate its ability to deliver voice-driven healthcare services in a realistic polyclinic setting. The voice interface was built using Streamlit, designed with elderly accessibility in mind, featuring large fonts and minimal dependence on touch-based interactions. On the backend, Supabase was used to handle real-time database interactions and enforce role-based access control (RBAC) for both patient and administrative operations. To demonstrate the system's practical viability, two web-based applications were deployed. Fig. 3a shows the first is a voice-enabled assistant interface for elderly users, which supports natural voice interaction to complete clinic tasks

such as appointment booking and billing inquiries. The second is a complementary admin portal for clinic staff, enabling backend operations such as patient record management and appointment updates, as shown in Fig. 3b and 3c. These applications simulate real-world workflows and demonstrate how *VoCare AI* can function as an integrated solution within a clinical setting.

IV. CONCLUSION

This paper presented *VoCare AI*, a voice-first conversational assistant that integrates a fine-tuned ASR model for Singapore-accented English, a LangGraph-based multi-agent orchestration framework, and a RAG module for grounded information retrieval. The system enables natural language interaction for core administrative tasks such as NRIC verification, appointment scheduling, and payment processing. Experimental results demonstrated strong performance in ASR and RAG on curated Singlish datasets, while end-to-end testing validated the system's robustness and usability in real-world clinical scenarios. Future work includes extending the system's multi-lingual capabilities to support Mandarin and Malay, enhancing intent disambiguation using dialogue memory embeddings, and integrating with national health platforms such as HealthHub and Singpass to enable secure, personalized services.

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